Building a User Simulator for Movie Recommendation

A Comparative Study of Static and Sequential Models

Predictive Minds
Bolai Yin · Junke Zhu · Peihan Wang

CS 6140 Machine Learning · Spring 2026 · Prof. Smruthi Mukund

Why Build a User Simulator?

The Problem

Offline RL for recommendation requires a reward signal, but real users cannot be queried on demand during training.

We build a user simulator that predicts **P(like | user, movie)**, enabling offline training and analysis for recommendation.

Our Research Question

Does the sequential order of a user's watch history carry meaningful predictive signal beyond aggregate statistics?

XGBoost (static) vs GRU4Rec (sequential)
Systematic ablation studies to isolate what matters.

Related Work: GRU4Rec (Hidasi et al., 2016) introduced session-based RNN for recommendations; XGBoost (Chen & Guestrin, 2016) dominates tabular prediction.

Dataset & Exploratory Analysis

1M	6,040	3,706	57.5%
Ratings	Users	Movies	Like Rate

Split Strategy

- Per-user temporal:

first 80% → train, next 10%→ validation, last 10% → test,

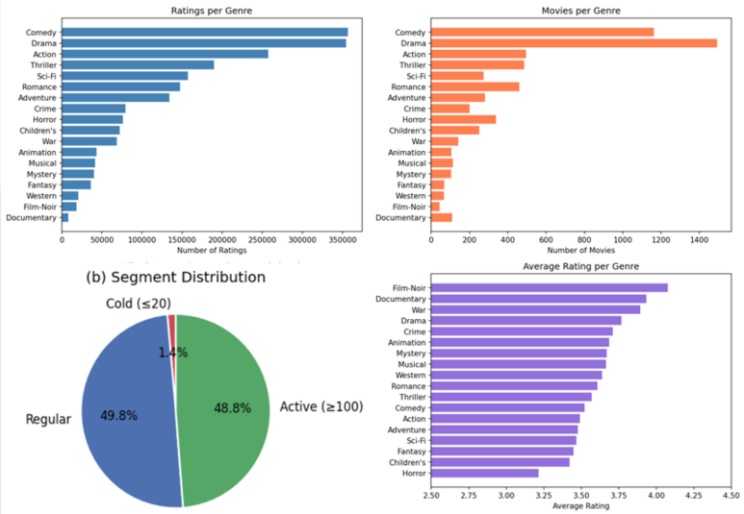
Train 797K | Val 100K | Test 103K | Zero data leakage verified

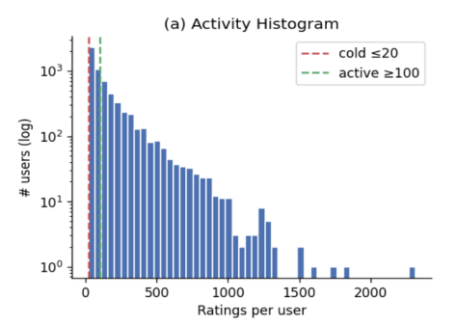
Why Per-User Split?

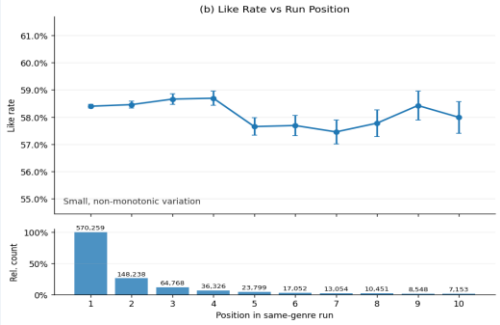
- Random split → user_id leakage (train-test gap 0.078).
- Per-user → gap only 0.013

Main takeaway

- MovieLens looks more preference-driven than session-driven.
- Fatigue effects are weak, which already hints that sequential gains may be limited.





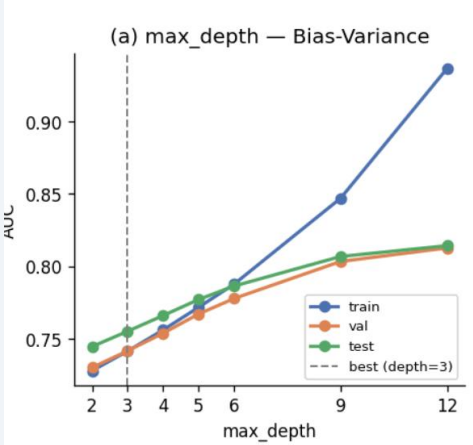


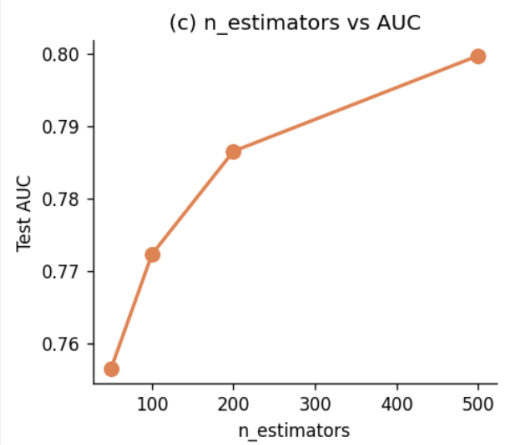
Model 1: XGBoost (Static)

Feature Group Contribution

Scheme	Dims	AUC	LogLoss
O: Static(meta + user demo + popularity)	24	0.7408	0.6015
A: SVD only	64	0.7995	0.5447
B: + movie meta	84	0.7997	0.5449
C: + user demographics	87	0.8003	0.5440
D: + popularity	88	0.7997	0.5449

→ User demographics: largest marginal gain
→ Popularity: near-zero gain — model uses personalization, not shortcuts





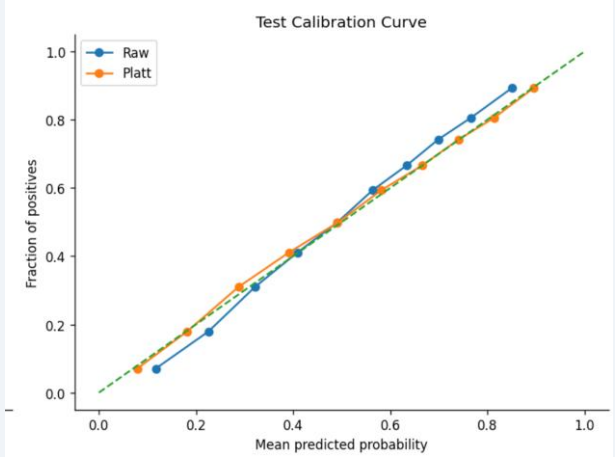
Hyperparams Tuning

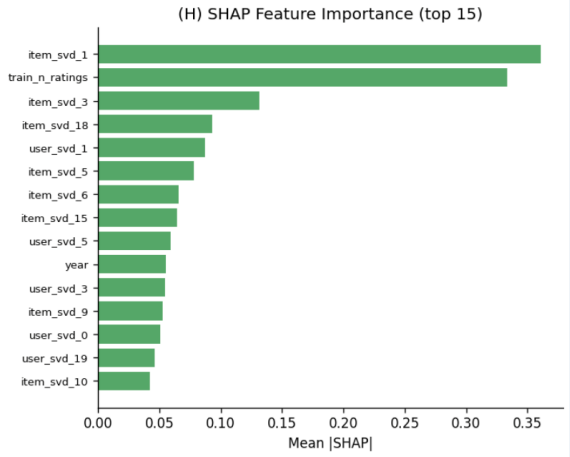
- n_estimators=500,
- max_depth=6,
- learning_rate=0.05,
- subsample=0.8,

XGBoost: Ablation & Results

Final Tuned XGBoost: Test AUC = 0.7997 LogLoss = 0.5447 F1 = 0.7391 (train-test gap = 0.016)

- *This is our final tuned XGBoost variant. It achieves about 0.80 test AUC with a relatively small train-test gap. Calibration improves log loss but does not change AUC, and recent similarity features do not help much. Overall, the result suggests that most of XGBoost's strength comes from latent collaborative features, while explicit popularity contributes very little marginal value.*





Key Ablation:

- Platt scaling $P(y=1|x)=\sigma(ax+b)$ lowers log loss, but AUC stays unchanged
- Recent similarity features do not help
- XGBoost is s driven mainly by latent SVD features and a small set of user and movie attributes; popularity alone is not the main driver.

Method 2: GRU4Rec (Sequential)

Architecture

Input: chronological sequence of user's watched movies + target movie
Output: probability of liking the target movie

Hyperparameters

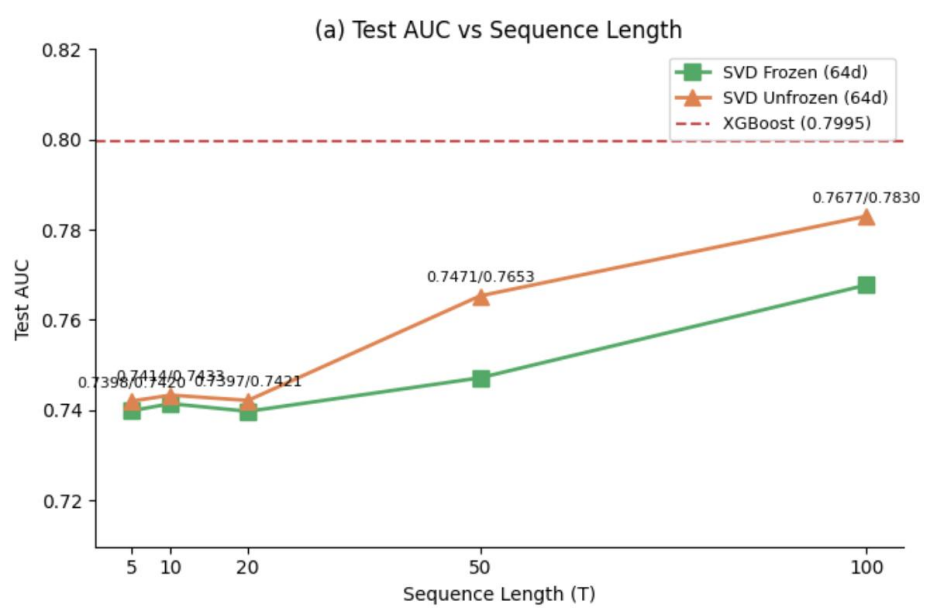
embed_dim: 64
hidden_dim: 64–512 (ablated)
seq_len T: 5–100 (ablated)
lr: 1e-3, dropout: 0.2
Early stopping (patience=5)
BCE loss, Adam optimizer

Best Params:
64d embedding + 128 hidden
T = 100

Takeaway
GRU4Rec models sequential preference directly from up to 100 past interactions.
Best GRU uses 1 layer: 2-layer GRU degrades substantially

GRU4Rec: Ablation Results

Sequence Length (T)

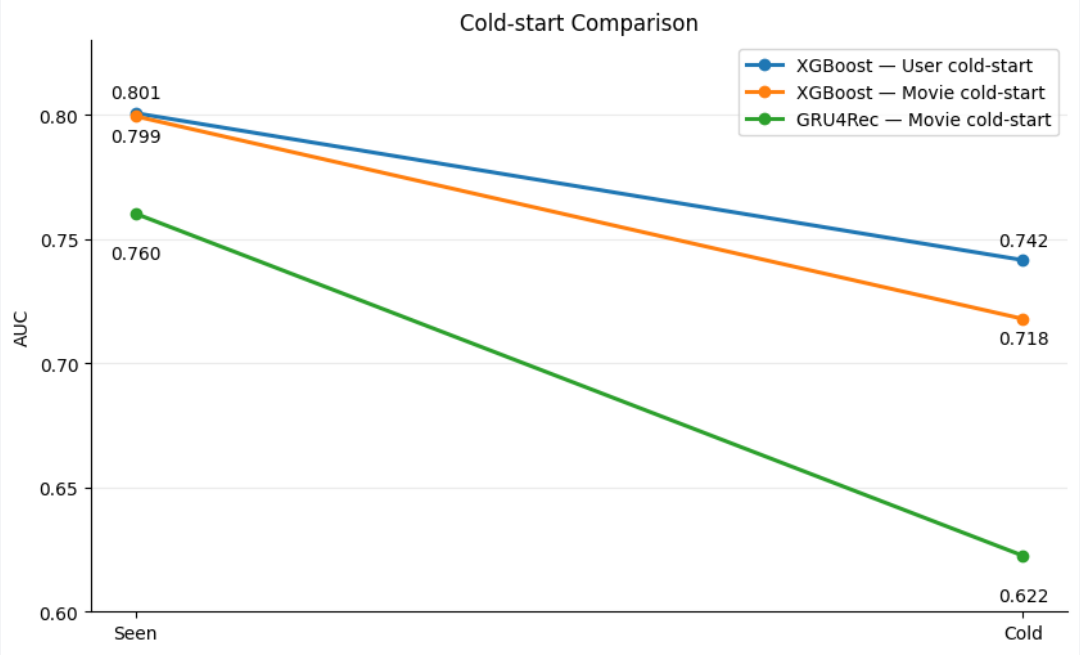


Best Params Comparison

Config	Hidden	T	Test - AUC
Learnable	128	100	0.7577
SVD-64 Frozen	128	100	0.7692
SVD-64 Unfrozen	128	100	**0.7830**

→ T=100 is best among tested values and already covers most of the history for a typical user (median = 96, mean = 165.6).

Cold-start comparison (XGB + GRU together)



Interpretation

- Strict holdout setup: held-out users/movies are removed entirely from training
- Both settings degrade, but performance remains above random
- Movie cold-start is harder than user cold-start in our SVD-based pipeline

→ Movie cold-start (balanced): hold out movies with 20–200 training interactions.
→ User cold-start: hold out users with at least 20 interactions.

Conclusions & Future Work

Conclusions:

- XGBoost is the best overall model on MovieLens 1M, reaching AUC ≈ 0.80
- Sequential gains are limited on this dataset: recent similarity features do not help XGBoost, and GRU4Rec remains below XGBoost
- Under strict holdout, movie cold-start is harder than user cold-start, and GRU4Rec is less robust than XGBoost on unseen movies.

Limitations

- MovieLens 1M shows weak temporal / sequential signal, which limits the upside of sequence models.
- GRU4Rec in the current implementation cannot naturally handle strict unseen users, since it requires observed interaction history.
- Unseen movies are represented by limited metadata only (mainly genre/year), which restricts cold-start generalization

Future Work

1. Better Datasets with stronger temporal dynamics
KuaiRec / Tenrec with richer sequential signals and natural fatigue patterns

2. Transformer Architectures
SASRec — attention may capture long-range dependencies better than GRU?

3. RL Integration
Use this simulator as the environment for training DQN/PPO recommendation agents (ongoing in CS5180)

4. Cold-start Mitigation
Explore content-based features or meta-learning to narrow the cold-user AUC gap (0.748 vs 0.809)

Team Contributions

Bolai Yin

Lead / XGBoost / Ablation Design

- Project architecture and code integration
- XGBoost modeling (features, training, tuning) and Ablation study design & analysis
- Cross-model comparison framework

Junke Zhu

GRU4Rec Implementation

- GRU4Rec model architecture (PyTorch)
- Sequential feature engineering
- Embedding ablation (learnable vs frozen SVD)
- Hidden dim and sequence length experiments
- Training loop and early stopping

Peihan Wang

Data & Experiment Tracking

- Data preprocessing pipeline
- Feature engineering (SVD, genre pooling)
- Experiment tracking and results recording
- Per-user temporal split implementation
- EDA visualizations

Shared tools: Google Colab (GPU training) · GitHub · MovieLens 1M · scikit-learn + XGBoost + PyTorch

Thank You

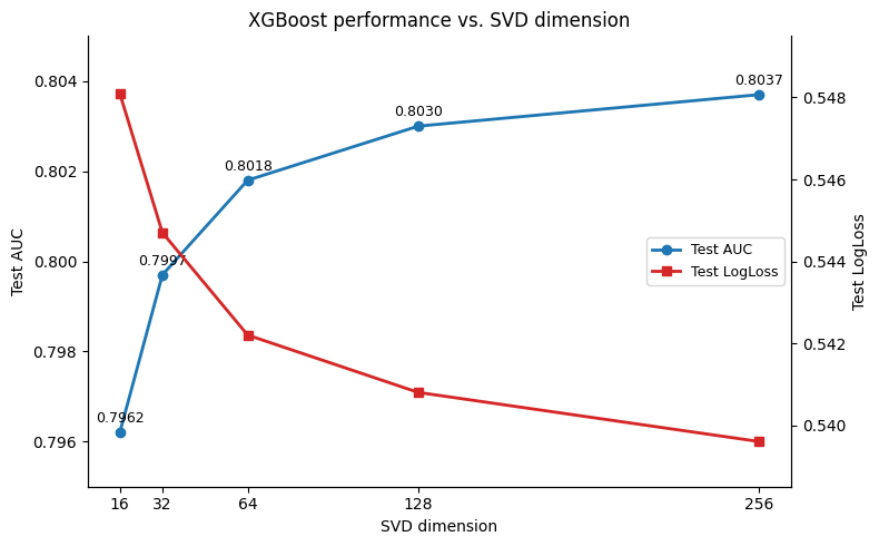
Questions?

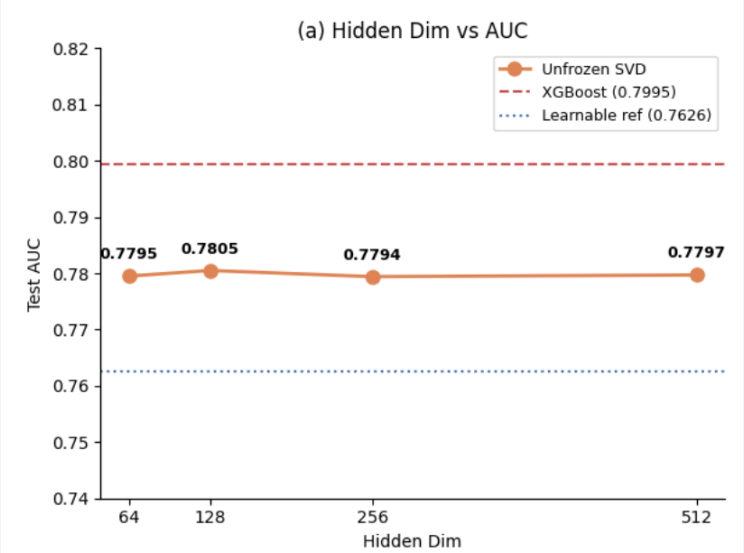
XGBoost AUC 0.800 · GRU4Rec AUC 0.770

Repo: github.com/Bryan-bits/CS6140-User-Simulator

Additional Ablation & Diagnostics

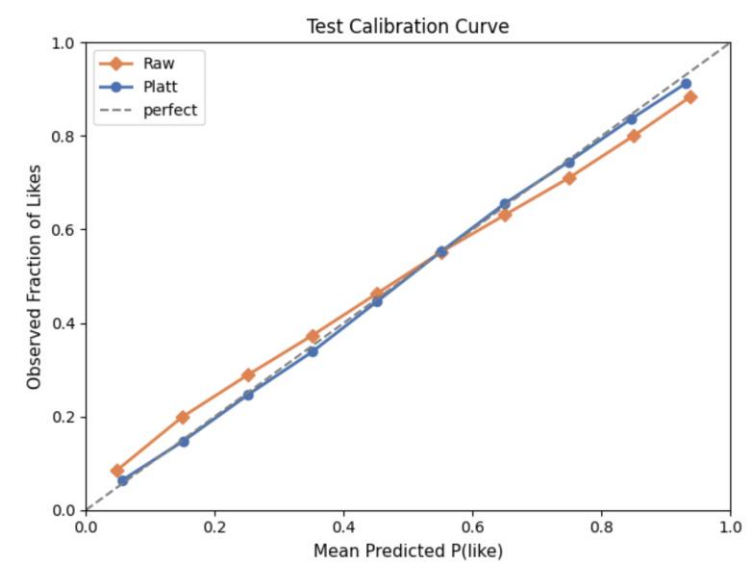
Final Tuned XGBoost: Test AUC = 0.7997 LogLoss = 0.5447 F1 = 0.7391 (train-test gap = 0.016)





Additional Ablation & Diagnostics

Final Tuned GRU4Rec: Platt LogLoss = 0.5592 ECE = 0.0313 -> 0.0064 (Δ = -0.0249)




```
Getting val predictions for Platt fitting...
  Val samples: 99,692
  Val AUC: 0.7835
  Platt params: a=0.8004, b=0.0246

GRU4Rec-SVD Unfrozen — Raw vs Platt Scaling

  Metric              Raw      Platt          Δ
  AUC              0.7825     0.7825    +0.0000
  LogLoss          0.5651     0.5592    -0.0059
  ECE              0.0313     0.0064    -0.0249
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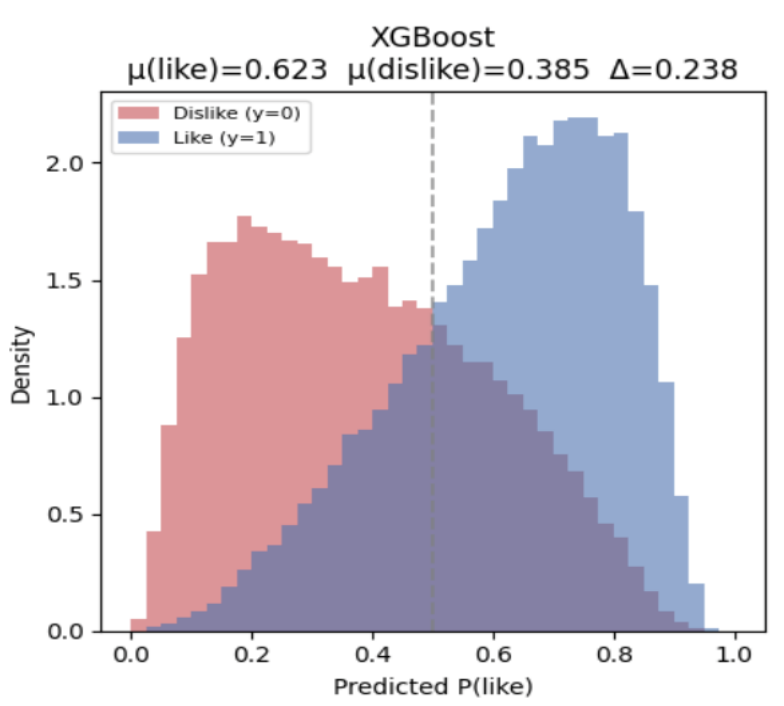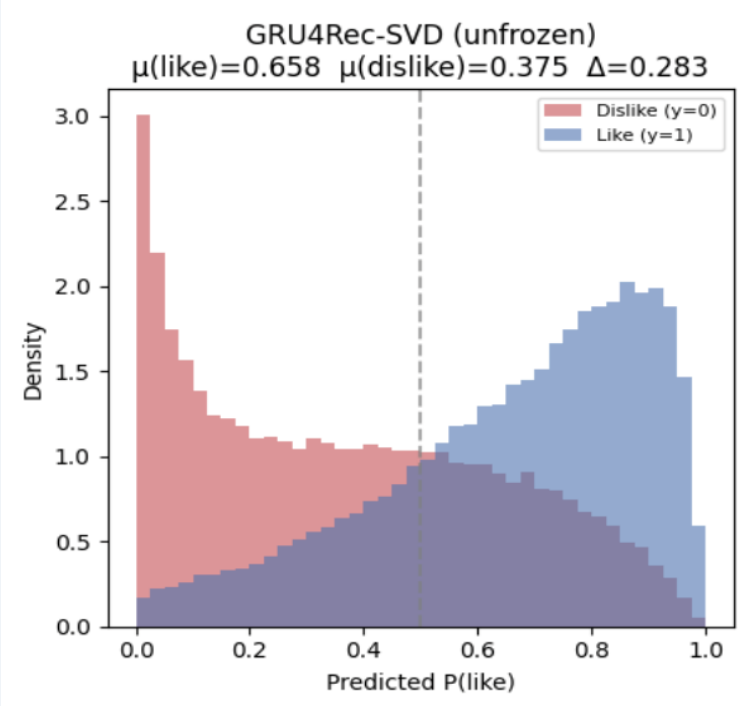
Additional Ablation & Diagnostics